Constructor University Bremen

MPFR LAB II

Part 4: The PID Controller For Speed Regulation
Spring Semester 2025

Task1.17 - Task1.23

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Task 1.17

First assume that the disturbace TL(s) = 0. Assume that a unit-step reference input r(t) = k u(t) is applied to the system, i.e. we want the controller to take the motor to a constant speed k rad/s. Apply the FVT to (1.32g) to find the steady state error e(t) = r(t) − ω(t).

$$T_L(s) = 0$$

$$r(t) = k\,u(t)$$

$$E(s) = \frac{1}{1 + H_{\omega v}(s)\,G(s)} R(s) + \frac{H_{\omega e}(s)}{1 + H_{\omega v}(s)\,G(s)} T_L(s)$$

$$= \frac{1}{1 + H_{\omega v}(s)\,G(s)} R(s) \,//$$

$$H_{\omega v}(s) = \frac{k_t}{D(s)}$$

$$= \frac{k_t}{(R_a + L_a s)(J_a s + b) + k_e k_t} \,//$$

$$G(s) = K_p + K_I \frac{1}{s} + K_D s$$

$$H_{\omega v}(s) \cdot G(s) = \frac{k_t}{(R_a + L_a s)(J_a s + b) + k_e k_t} \cdot \left(K_p + K_I \frac{1}{s} + K_D s\right) \cdot \frac{s}{s}$$

$$= \frac{k_t(K_p s + k_I + k_D s^2)}{s(R_a + L_a s)(J_a s + b) + k_e k_t}$$

$$1 + A_{ns} = \frac{s(R_a + L_a s)(J_a s + b) + k_e k_t + k_t(K_p s + K_I + K_D s^2)}{s(R_a + L_a s)(J_a s + b) + k_e k_t}$$

$$R(s) = k\,u(t) = \frac{k}{s}$$

$$E(s) = \lim_{s \to 0} \frac{s(R_a + L_a s)(J_a s + b) + k_e k_t}{s(R_a + L_a s)(J_a s + b) + k_e k_t + k_t(K_p s + K_I + K_D s^2)} \frac{k}{s}$$

$$= 0 \,//$$

Task 1.18.

Next assume that r(t) = k u(t) and TL(t) = ` u(t), i.e. the system is now also disturbed by a load-torque ` Nm. Using the FVT, determine the effect of this disturbance on the system steady-state error.

$$r(t) = k\,u(t)$$

$$T_L(s) = \ell\,u(t) \quad , \quad = \frac{\ell}{s}$$

$$E(s) = \underbrace{\frac{1}{1 + H_{wr}(s)\,G(s)}\,R(s)}_{\text{From Task 1.17} = 0} + \frac{H_{we}(s)}{1 + H_{wr}(s)\,G(s)}\,T_L(s)$$

$$= \frac{H_{we}(s)}{1 + H_{wr}(s)\,G(s)}\,T_L(s)$$

$$H_{we}(s) = \frac{R_a + L_a s}{D(s)}$$

$$= \frac{R_a + L_a s}{(R_a + L_a s)(J_a s + b) + k_e k_t}$$

From Task 1.17,

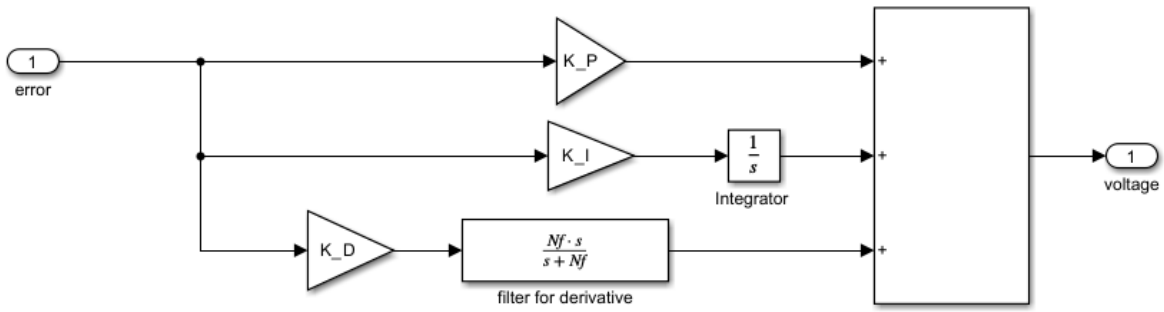
$$1 + H_{wr}(s)\,G(s) = \frac{s(R_a + L_a s)(J_a s + b) + k_e k_t + k_t(K_p s + K_I + K_D s^2)}{s(R_a + L_a s)(J_a s + b) + k_e k_t}$$

$$E(s) = \lim_{s \to 0} \frac{R_a + L_a s}{(R_a + L_a s)(J_a s + b) + k_e k_t} \; \frac{s(R_a + L_a s)(J_a s + b) + k_e k_t}{s(R_a + L_a s)(J_a s + b) + k_e k_t + k_t(K_p s + K_I + K_D s^2)} \; \frac{\ell}{s}$$

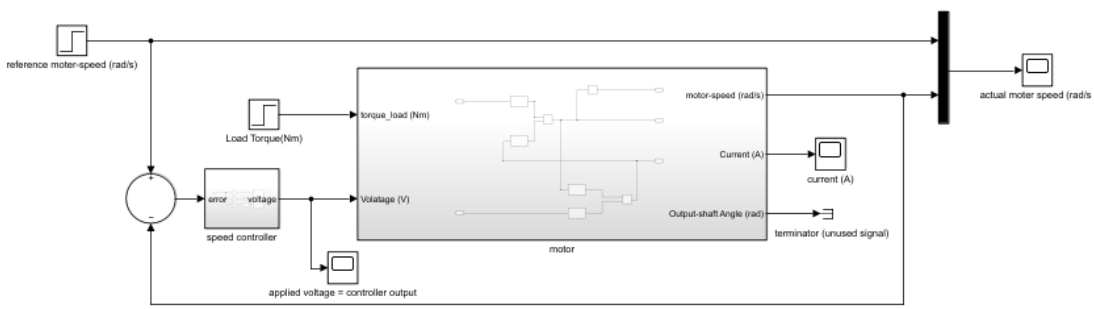
$$= 0$$

Task 1.19.

Copy your previous motor-model .slx file to another file and open it in Simulink. In this new model, create the controller subsystem connected to the motor as shown in Fig. 1.12.



Above is the speed controller, with Nf = 100, K_P = 0.0084, K_I = 0.02, K_D = 0

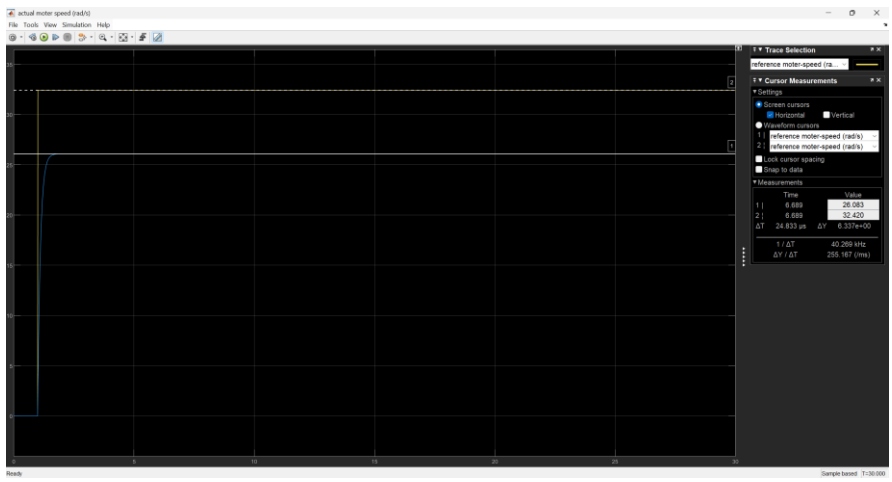


Task 1.20.

1. *In our controller, let us set KI = KD = 0 and KP = 1. Let us select a reference speed r(t) = 300 RPM (convert to rad/s before applying to the model) which is less than ω0 and hence doable. Set the load-torque input to 0. Look at the scope: Did ω(t) reach r(t)? If not, measure the steady-state error e(t).*

2. *Since there is some steady-state error, let us increase KP to 10 and simulate again. What is the steady-state error now? Since it decreased, we could just keep increasing KP till the steady-state error is smaller than some threshold, right?*

3. *Wrong. Some part of system just went berserk at KP = 10. Look at the scopes to figure out what went wrong. The moral of the story is that if we just use KP, there will always be a steady-state error. To have it converge to zero, we need a non-zero KI.*

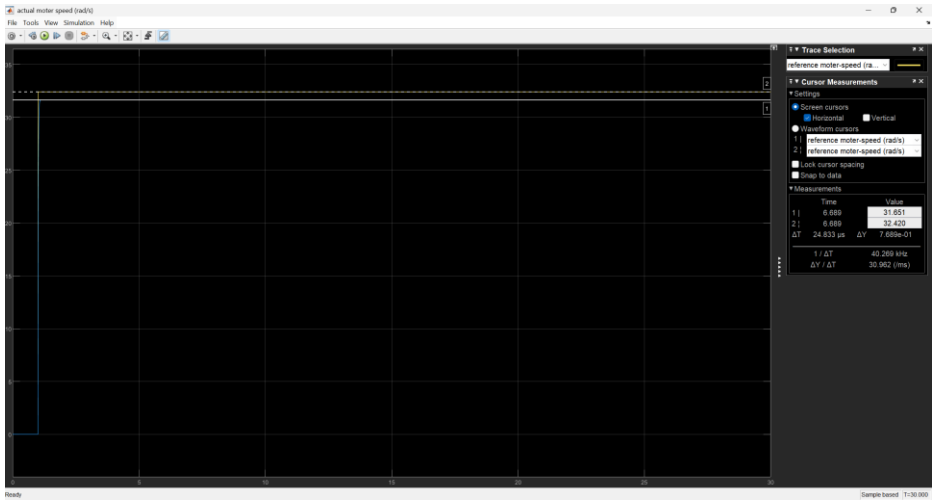


Above is the speed controller, with changed

K_I = K_D = 0, K_P = 1, load-torque input to 0, r(t) = 300RPM≈31.42 rad/sec

Since ω(t) didn't reach r(t),

the steady state error = e(t) = r(t) - ω(t) = 32.420 – 26.83 = 5.59



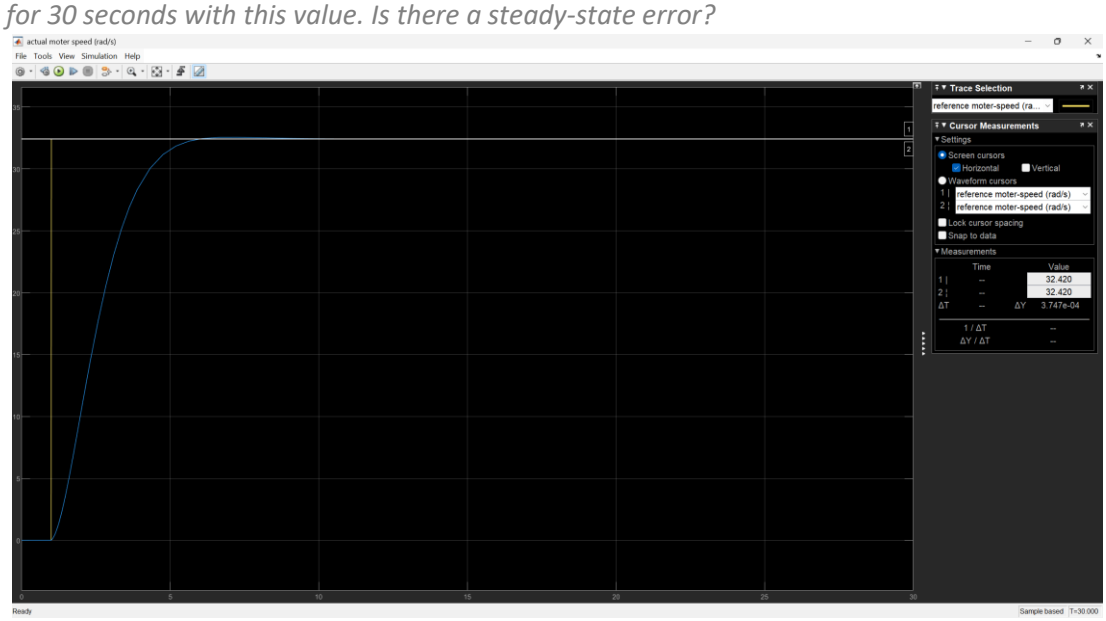
Above increased K_P to 10.

the steady state error = e(t) = r(t) - ω(t) = 32.420 – 31.651 = 0.769

Task 1.21.

Proceed as follows:

1. *Let us set KD = 0 as before but set KP = 0.0084. Select a value of KI where the system is stable but just begins to show some oscillations. You can find this by writing an expression for the two roots of (1.37) in terms of KI and find the value of KI when the discriminant just turns 0. Simulate the system for 30 seconds with this value. Is there a steady-state error?*

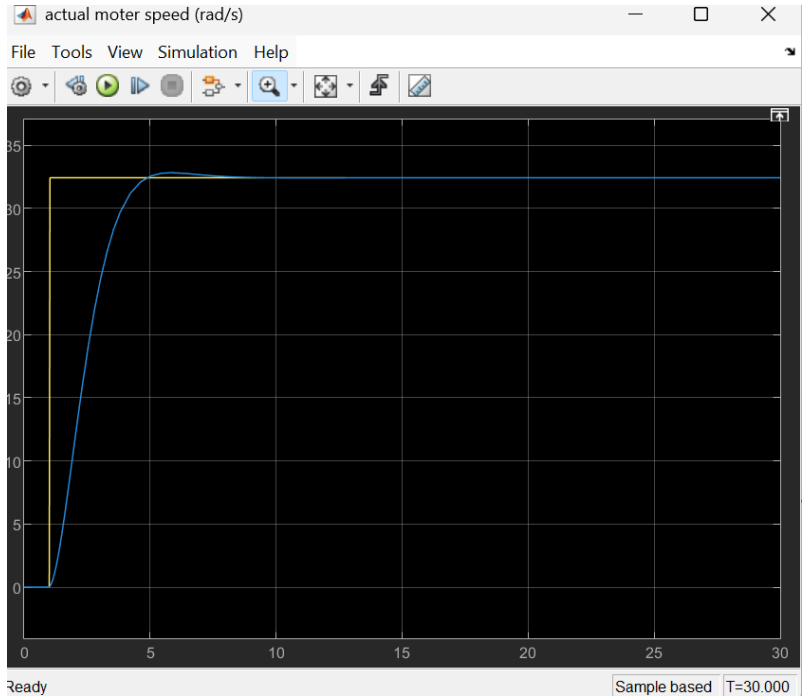


Above is the speed controller, with changed

K_D = 0, K_P = *0.0084*, K_I = 0.15

There is no steady-state error.

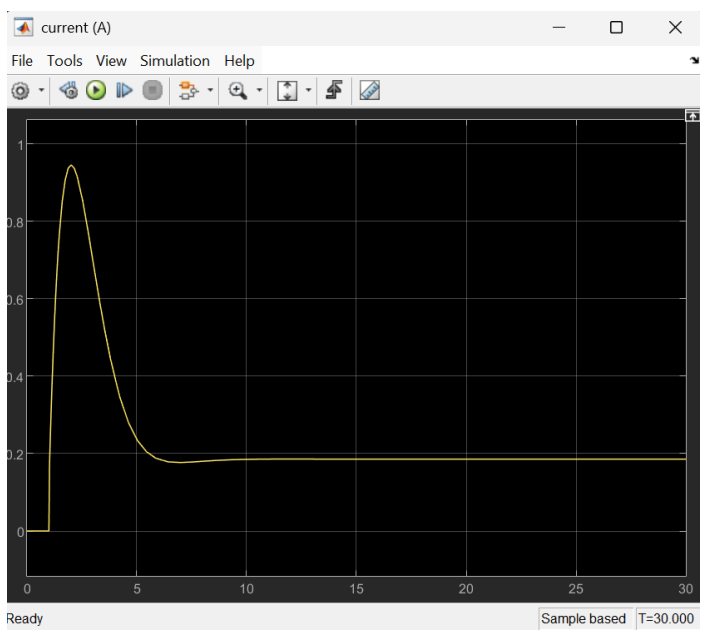
2. *Now select a value of KI slightly above this threshold, and simulate again. Do you see a ripple before the system settles down?*



Slightly adjust the K_P = 0.17

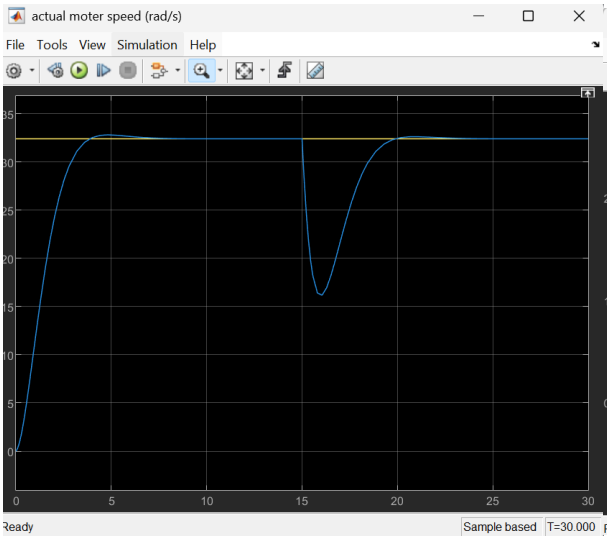
There is a ripple before it becomes steady.

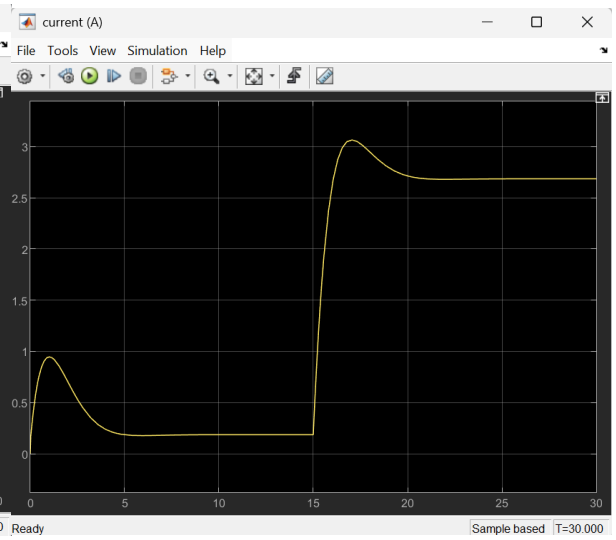
3. *Now look at the current values in Amperes. Are they in the safe range? The nice thing about our controller is that it even damps the current-spike when a step-input is applied.*



It is in safe range below 5.

4. *Now add a step load-torque disturbance which is half of the stall torque at t = 15 seconds. Did the steady-state error still go to zero? Also look at the other scopes: Are all values within limits? Refer to Fig. 1.13 for an example.*



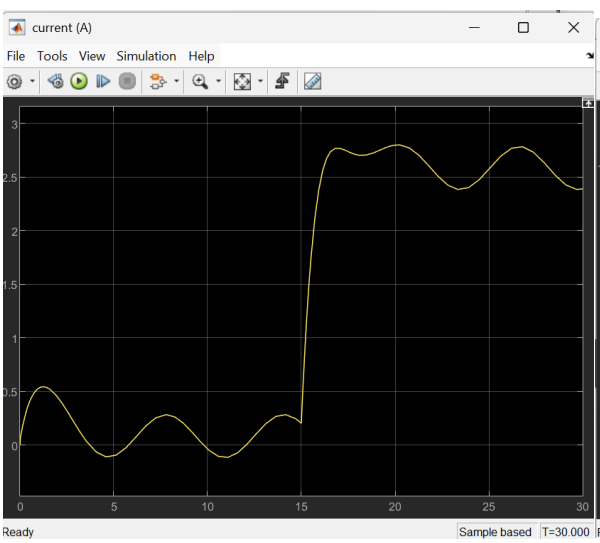


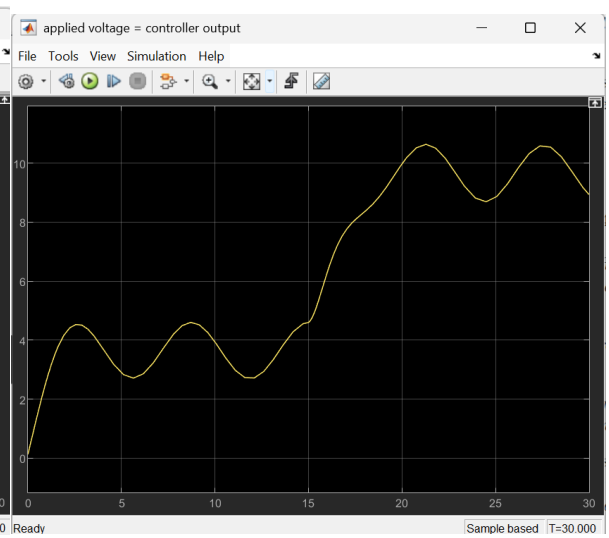
Above set load torque to t = 15s, and stall torque = 0.5932 /2 = 0.2966

The steady state error is still zero.

All the values are still within the limits, below 5.

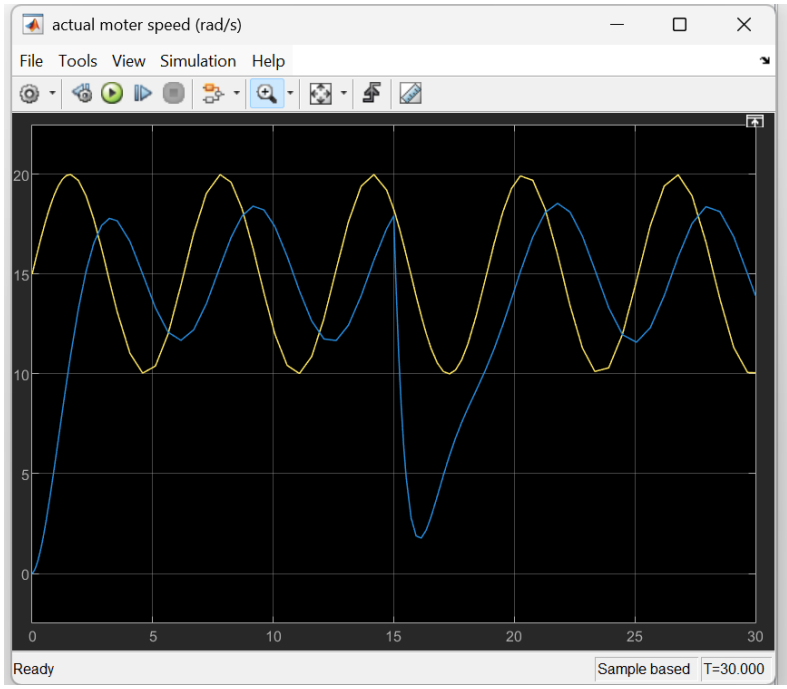
5. *You are, of course, not restricted to step-inputs. Let us apply a reference-speed signal which is sinusoidal with amplitude of 5 rad/s and a bias of 15 rad/s and the frequency of the sine-wave is 1 rad/s. Keep the step torque-load at t = 15 sec. Your scope should look like Fig. 1.14. Are the voltages and currents within limits?*





They are within limits, in which voltage is below 12V and current below 5A.

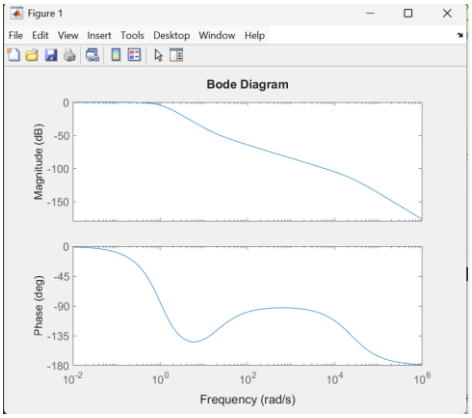
6. In the previous example, let's try to see if the observed amplitude reduction and phase difference between the reference and the actual speed sinusoidal signals can be also predicted by the Bode plot. The Bode plot of which transfer function will you have to plot?



Error , E(t)

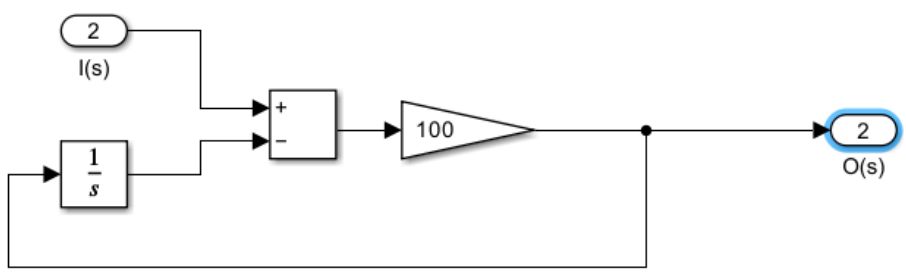
Task 1.22.

To plot the Bode plot proceed as shown in the code listing below. Make sure you use your chosen controller constants rather than the ones shown. Do the magnitude and phasedifference shown by the Bode plot at 1 rad/s match with what you observe in the speed scope?




```
>> bode(T)
>> grid on;
>> w =1;
>> [mag,phase] = bode(T,w)

mag =

    0.5974


phase =

  -84.8818
```

Task 1.23.

How will you realize the derivative filter using a gain (multiplication by a constant) block and negative feedback with an integrator block? Hint: On the Matlab command line try Hsp= feedback (Nf, 1/s, -1). Now draw this as a block diagram using only gain, integrator and a negative feedback.

```
>> Hsp= feedback(Nf, 1/s, -1)

Hsp =

   100 s
  -------
  s + 100

Continuous-time transfer function.
Model Properties
```